

Explaining Sensor Measurements for Informing Peatland Restoration and Conservation

Martin Aleksandrov, Florian Jansen, Anke Günther

University of Rostock, Germany

Abstract

We present a novel AI approach for explaining sensor measurements on peatlands. Over the second half of 2025, we designed and implemented an unprecedented sensor network across more than 100 peatland screening sites in North-East Germany. Sensors in this network quantify air, soil, groundwater, and evapotranspiration physical properties, and send their measured quantities as raw-data messages through a LoRaWAN network to a time-series InfluxDB instance hosted at our university. Our explainability approach has the following steps: 1) we conduct an observation-level evaluation of the raw-data points [e.g., observation tests, flag assignment, anomaly detection]; 2) based on these evaluations, we correct/calibrate of the raw-data points [e.g., drift correction, calibration offset, gap detection]; and, finally, 3) we estimate the quality of a set of data points over a fixed period of time [e.g., completeness, validity, latency, coverage]. Additionally, we propose two indices -- the Sensor Quality Index (SQI) and the Dataset Quality Index (DQI) -- and compute them for different sensors and the datasets flowing from sensors at our screening sites. Based on these indices, we rank sensors and datasets to classify them as "good", "suspicious", and "bad". Our approach generates explanation texts justifying the quality of sensors and datasets. This supports policy-making for sensor deployment. For example, suspicious and bad sensors can be re-set up or potentially replaced. Good sensors can be used to cluster the sites into groups: e.g., sites where sensors are "good" and produce similar measurements fall into the same group. When combined with chemical soil and groundwater properties, our approach informs restoration and conservation as similar sites should be treated similarly, which facilitates the long-term management of these activities.